IgM (contains μ chain), IgD (contains δ chain) and IgE (contains ε chain). Five heavy chain classes of immunoglobulin can be easily remembered as GAMDE. Heavy chain classes are again subdivided into subclasses of molecules.

i. Four known subclasses of the γ chain exist – γ1, γ2, γ3 and γ4 - which yield IgG1, IgG2, IgG3 and IgG4.

ii. Two subclasses of the α chain are known – α1 and α2 - which yield IgA1 and IgA2.

iii. Two subclasses of the μ chain are known – μ1 and μ2 - which yield IgM1 and IgM2.

iv. No subclasses of the δ and ε (IgD and IgE) are known.

2. Disulfide bonds

Disulfide bonds hold together the four polypeptide chains in normal immunoglobulin molecules and are of two types namely interchain bonds and intrachain bonds.

3. Regions

Each heavy and light chain consists of two segments, the variable region and the constant region. The variable (V) region shows a wide variation in amino acid sequence in the amino terminal portion of the molecule.

4. Domains

Each immunoglobulin chain consists of a series of globular regions enclosed by disulphide bonds. Each heavy chain consists of four or five domains - one in the variable region (VH) and three or four in the constant region (CH1, CH2, CH3, and CH4). Each light chain consists of two domains – one in the variable region (VL)and one in the constant region (CL).

5. Fragments.

Proteolytic (peptide bond -splitting) enzymes such as papain and pepsin are used to degrade immunoglobulin molecules into definable fragments to facilitate study of their structure Figure (13.22).

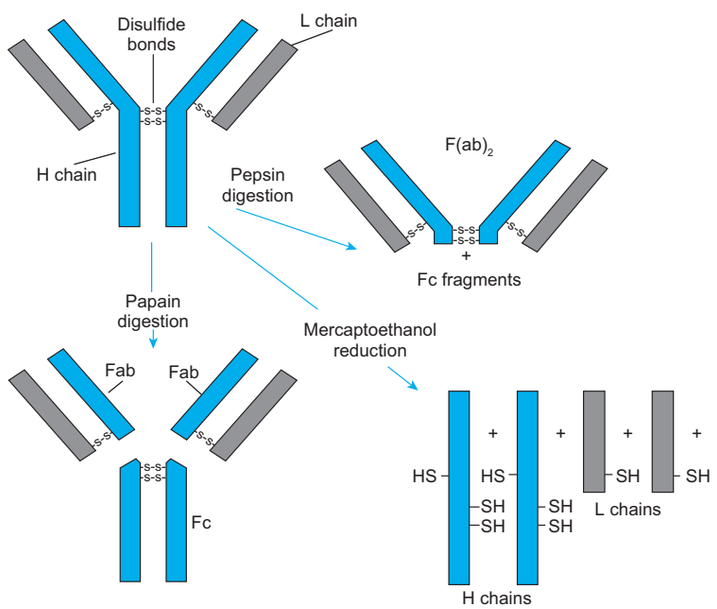

Figure 13.22: Prototype structure of IgG, showing chain structure and interchain disulfide bonds

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